Distributed Agent Registry — Attribute Specification

Overview

A Distributed Agent Registry (DAR) provides a standardized way for AI agents to prove their identity, capabilities, authorization scope, and payment readiness to merchants and service providers.

This design synthesizes three industry approaches:

- **Visa TAP**: Cryptographic agent identity + intent signatures
- **Google UCP/AP2**: Capability discovery + payment handler interop
- **Skyfire KYA**: Know-Your-Agent verification via JWT

The registry is distributed — no single authority. Multiple issuers (PSPs, platforms, identity providers) can register agents. Merchants verify agents using public keys fetched from the issuer's JWKS endpoint.

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1. Agent Identity Attributes

Core identity established during boarding. Immutable after issuance.

```yaml
```

```yaml
agent_identity:
  # ── Required ──────────────────────────────────────────────
  agent_id: "agent_a1b2c3d4e5f6"      # Globally unique
  agent_did: "did:web:psp.example.com:agents:a1b2c3d4e5f6"  # W3C DID
  name: "Alpha Trading Agent"
  version: "1.2.0"                     # Semantic version of agent software

  # Who owns/operates this agent
  owner:
    entity_id: "org_vinit_labs"
    name: "Vinit Labs Inc."
    contact: "vinit@example.com"
    jurisdiction: "US"                 # ISO 3166-1 alpha-2

  # Who issued this identity (the registry authority)
  issuer:
    issuer_id: "iss_psp_example"
    name: "Example PSP"
    jwks_uri: "https://psp.example.com/.well-known/jwks.json"
    trust_framework: "visa-tap"        # visa-tap | google-ap2 | skyfire-kya | cust

  # ── Optional ──────────────────────────────────────────────
  platform: "openai"                   # Agent runtime platform
  model: "gpt-4o"                      # Underlying model (if applicable)
  homepage: "https://agent.example.com"
  logo_uri: "https://agent.example.com/logo.png"
```

2. Agent Capabilities & Scope

What the agent is authorized to do. Defines the trust boundary.

```yaml
```

```yaml
capabilities:
  # ── Intent Types (what the agent can do) ──────────────────
  intents:
    - browse           # View product/service details
    - purchase         # Execute transactions
    - subscribe        # Create recurring payments
    - return           # Initiate returns/refunds
    - negotiate        # B2B price negotiation

  # ── Commerce Capabilities (per Google UCP model) ──────────
  commerce:
    protocols_supported:
      - "mpp"          # Machine Payments Protocol
      - "ucp"          # Google Universal Commerce Protocol
      - "acp"          # OpenAI Agentic Commerce Protocol
      - "x402"         # Coinbase x402
      - "ap2"          # Google Agent Payments Protocol
    transport:
      - "http"         # Direct HTTP APIs
      - "mcp"          # Model Context Protocol
      - "a2a"          # Google Agent-to-Agent

  # ── Spending Limits ───────────────────────────────────────
  spending:
    per_transaction_limit: "10000"      # Base units ($100.00)
    daily_limit: "500000"               # $5,000/day
    monthly_limit: "5000000"            # $50,000/month
    currencies: ["usd", "eur", "gbp"]
    auto_approve_threshold: "1000"      # Below this: no human approval needed

  # ── Category Restrictions ─────────────────────────────────
  allowed_categories:
    - "digital_services"
    - "saas"
    - "api_access"
    - "compute"
    - "data_feeds"
  blocked_categories:
    - "gambling"
    - "adult_content"
    - "weapons"

  # ── Geographic Restrictions ───────────────────────────────
```

```yaml
  geo:
    allowed_countries: ["US", "GB", "EU"]  # ISO 3166
    blocked_countries: ["KP", "IR"]
```

3. Consumer / Delegation Context

Ties the agent back to the human or organization it acts for.
Matches Visa TAP's "Consumer Recognition" layer.

```yaml
delegation:
  # ── Who the agent acts on behalf of ──────────────────────
  consumer:
    consumer_id: "usr_vinit_42"
    consumer_did: "did:key:z6MkhaXgBZDvotDkL5257faiztiGiC2QtKLGpbnnEGta2doK"
    verified: true                      # Identity provider has verified
    identity_provider: "auth0"

  # ── Consent / Authorization ──────────────────────────────
  consent:
    type: "delegated"                   # delegated | autonomous | supervised
    scope: "purchase:digital_services"
    granted_at: "2026-04-01T00:00:00Z"
    expires_at: "2026-07-01T00:00:00Z"
    revocation_uri: "https://psp.example.com/consent/revoke/abc123"

  # ── Merchant Recognition (per Visa TAP) ──────────────────
  merchant_context:
    loyalty_id: "LOY-12345"             # If consumer has merchant account
    device_fingerprint: "fp_abc123"     # For returning customer detection
    prior_interaction: true             # Has consumer bought before?
    location:
      country: "US"
      postal_code: "94105"
```

4. Payment Readiness

What payment methods the agent can use. Matches Visa TAP's
"Payment Information" layer and MPP's payment method model.

```yaml
```

```yaml
payment:
  # ── Wallet Reference ──────────────────────────────────────
  wallet_id: "wallet_x1y2z3a4b5c6"
  psp_id: "psp_example"
  psp_endpoint: "https://psp.example.com"

  # ── Available Payment Methods ─────────────────────────────
  methods:
    - method_id: "pm_visa_4242"
      type: "card"
      network: "visa"
      last4: "4242"
      expiry: "12/28"
      tokenized: true                    # Network token, not raw PAN
      payment_account_reference: "PAR-V-abc123"  # Visa PAR for card-on-file

    - method_id: "pm_usdc_wallet"
      type: "stablecoin"
      network: "tempo"
      address: "0x742d35Cc6634C0532925a3b844Bc9e7595f8fE00"
      currency: "USDC"

    - method_id: "pm_lightning"
      type: "lightning"
      network: "bitcoin"
      node_pubkey: "03abc..."

  # ── Active Tokens ─────────────────────────────────────────
  active_tokens:
    single_use: 3                        # tok_ tokens currently active
    multi_use: 1                         # mtok_ sessions open
    total_held: "100000"                 # $1000 held across sessions
    total_available: "400000"            # $4000 remaining spending capacity
```

5. Cryptographic Identity (per Visa TAP)

Agent signs every request. Merchant verifies using issuer's public key.

```yaml
```

```yaml
cryptography:
  # ── Signing Key ──────────────────────────────────────────
  algorithm: "ed25519"                  # ed25519 | ecdsa-p256 | rsa-pss
  key_id: "kid_agent_a1b2c3_2026Q2"    # Rotated quarterly
  public_key: "z6MkhaXgBZDvotDkL5257faiztiGiC2QtKLGpbnnEGta2doK"

  # ── Signature Components (per HTTP Message Signatures RFC) ─
  signature_components:
    - "@authority"                       # Merchant domain
    - "@path"                           # Resource being accessed
    - "@method"                         # GET, POST, etc.
    - "signature-agent"                 # Agent DID
    - "created"                         # Timestamp
    - "nonce"                           # Replay protection
    - "tag"                             # "visa-tap" or "mpp"

  # ── Verification ─────────────────────────────────────────
  jwks_uri: "https://psp.example.com/.well-known/jwks.json"
  certificate_chain: "https://psp.example.com/certs/agent_a1b2c3.pem"
```

6. Trust & Compliance

Audit trail, risk signals, compliance status.

```yaml
```

```yaml
trust:
  # ── Trust Score (computed by registry/PSP) ──────────────
  trust_score: 92                       # 0-100, based on history
  trust_level: "verified"               # unverified | basic | verified | premium

  # ── KYA (Know Your Agent) Status ────────────────────────
  kya:
    status: "approved"                  # pending | approved | suspended | revoked
    verified_at: "2026-03-15T10:00:00Z"
    next_review: "2026-06-15T00:00:00Z"
    checks_passed:
      - "identity_verification"
      - "owner_kyb"                     # Know Your Business on owner
      - "spending_pattern_analysis"
      - "fraud_screening"

  # ── Compliance ──────────────────────────────────────────
  compliance:
    psd2_sca_exempt: false              # EU Strong Customer Auth
    pci_dss_compliant: true
    gdpr_data_processing: "compliant"
    aml_screening: "passed"

  # ── Audit Trail ─────────────────────────────────────────
  audit:
    total_transactions: 14523
    total_volume: "2847200"             # $28,472.00
    first_transaction: "2026-01-15T08:00:00Z"
    last_transaction: "2026-04-03T21:30:00Z"
    dispute_rate: "0.001"               # 0.1%
    chargeback_count: 2
```

7. Registry Metadata

How the registry entry itself is managed.

yaml

```yaml
registry:
  # ── Entry Lifecycle ──────────────────────────────────────────
  entry_id: "dar_entry_a1b2c3d4e5f6"
  status: "active"                      # active | suspended | revoked | expired
  created_at: "2026-01-10T00:00:00Z"
  updated_at: "2026-04-03T21:00:00Z"
  expires_at: "2027-01-10T00:00:00Z"

  # ── Distribution ─────────────────────────────────────────────
  registry_uri: "https://psp.example.com/agents/a1b2c3d4e5f6"
  discovery_endpoint: "https://psp.example.com/.well-known/agent-registry"

  # ── Revocation ───────────────────────────────────────────────
  revocation:
    crl_uri: "https://psp.example.com/crl"
    ocsp_uri: "https://psp.example.com/ocsp"

  # ── Cross-Registry Federation ────────────────────────────────
  federated_registries:
    - registry: "visa-tap"
      external_id: "visa_agent_xyz"
      status: "linked"
    - registry: "google-ap2"
      external_id: "ap2_agent_abc"
      status: "linked"
    - registry: "skyfire-kya"
      external_id: "kya_agent_def"
      status: "pending"
```

How It Fits Together

```
┌──────────────────────────────────────────────────────────────┐
│                  Distributed Agent Registry                  │
│                                                              │
│  ┌──────────┐  ┌──────────┐  ┌──────────┐  ┌──────────┐    │
│  │ Identity │  │Capability│  │Delegation│  │Payment   │    │
│  │ agent_id │  │ intents  │  │ consumer │  │ wallet   │    │
│  │ DID      │  │ spending │  │ consent  │  │ cards    │    │
│  │ owner    │  │ geo/cats │  │ loyalty  │  │ tokens   │    │
│  │ issuer   │  │ protocols│  │ location │  │ crypto   │    │
│  └──────────┘  └──────────┘  └──────────┘  └──────────┘    │
```

```
Crypto Identity
signing key
HTTP signatures
JWKS endpoint

Trust & Compliance
KYA status
trust score
audit trail
dispute rate

Registry Metadata
lifecycle, discovery, revocation, federation

Visa TAP Link
(federated)

Google AP2 Link
(federated)

Skyfire KYA Link
(federated)
```

Key Design Decisions

1. **DID-based identity** — W3C Decentralized Identifiers allow agents to have portable identity across registries without a central authority.
2. **Federated, not centralized** — Multiple PSPs/platforms can issue agent entries. Cross-registry linking allows a single agent to be recognized across Visa TAP, Google AP2, and Skyfire KYA simultaneously.
3. **HTTP Message Signatures** — Following Visa TAP's approach, agents sign every HTTP request. Merchants verify using the issuer's JWKS endpoint. No custom crypto — standard RFC 9421 signatures.
4. **Separation of identity and payment** — An agent's identity is independent of its payment methods. The same agent can have cards, stablecoins, and Lightning — the registry tracks all of them.
5. **Trust is earned, not declared** — Trust score is computed from transaction history, dispute rate, KYA checks, and compliance status. New agents start at "basic" and earn "verified" through clean history.
6. **Consent is explicit and revocable** — Every delegation has a scope, expiry, and revocation URI. Matches PSD2/SCA requirements for the EU.